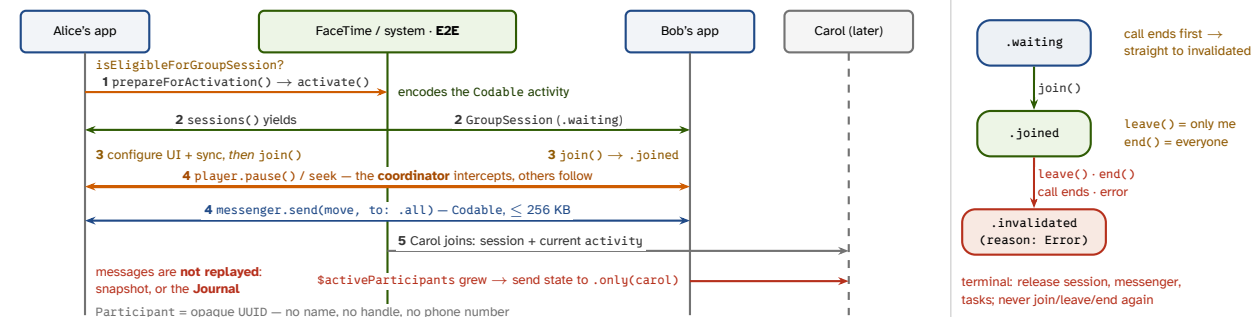


In one line: **SharePlay** = your app's activity shared over a **FaceTime** call or **Messages** conversation (and, iOS 17, **AirDrop** nearby). You define a Codable GroupActivity; one person activate()s it; the system hands a GroupSession to every participant's sessions() stream; each app join()s and syncs through **AVPlayer's playback coordinator** (media), **GroupSessionMessenger** (small messages) or **GroupSessionJournal** (files). All session data is **end-to-end encrypted**; there is no server of yours in the loop.

One FaceTime call, three devices — who calls what



How it works

- **Capability:** *Group Activities* → entitlement `com.apple.developer.group-session`. Framework since iOS 15 / macOS 12 / tvOS 15 / visionOS 1.
- **Activity** = struct: `GroupActivity` (Codable, sent to every device) with static `activityIdentifier` (needed once you have several types) and metadata: `title`, `subtitle`, `previewImage` (CGImage), `fallbackURL` (join from a web browser), `type` (`.watchTogether`, `.listenTogether`, `.playTogether`, `.createTogether`, ... iOS 17 added `shop/read/learn/workout`), `supportsContinuationOnTV`.
- **Start:** `GroupStateObserver().isEligibleForGroupSession` (ObservableObject) — true in a call/conversation. `prepareForActivation()` asks the user: `.activationPreferred` → `activate()` · `.activationDisabled` → do it *locally* · `.cancelled` → nothing. Not eligible → present `GroupActivitySharingController` (iOS 15.4): it invites people and starts the call.
- **Share sheet / AirDrop:** `ShareLink(item: activity, ...)` with a Transferable activity; UIKit: `NSItemProvider.registerGroupActivity(_:)` (15.4) in `UIActivityViewController` or a responder's `activityItemsConfiguration` — the latter is what **SharePlay over AirDrop** (iOS 17) finds.
- **Receive** on every device, including the starter: for await `s in X.sessions()` from launch (e.g. `.task` on the root view). Store it, build UI + sync objects, *then* `join()`.
- **Change content:** set `session.activity` (@Published, only when `.joined`); everyone observes `$activity` and loads the same item.
- Also on the session: `isLocallyInitiated` (am I the starter?), `requestForegroundPresentation()` (the activity needs your app on screen).

Example — watch together + reactions

```

struct WatchTogether: GroupActivity {
    let movie: Movie // Codable payload
    var metadata: GroupActivityMetadata {
        var m = GroupActivityMetadata(); m.type = watchTogether
        m.title = movie.title; m.fallbackURL = movie.webURL; return m
    }
}

func start(_ a: WatchTogether) async throws { // Alice's tap
    switch await a.prepareForActivation() {
    case .activationPreferred: _ = try await a.activate()
    case .activationDisabled: playLocally(a.movie)
    default: break } // .cancelled
}

for await s in WatchTogether.sessions() { // every device
    session = s; load(s.activity.movie) // keep strong refs
    player.playbackCoordinator.coordinateWithSession(s)
    messenger = GroupSessionMessenger(session: s, deliveryMode: .unreliable)
    s.join()
    Task { for await r, ctx in messenger.messages(of: Reaction.self) {
        show(r, from: ctx.source) } } // ctx.source: Participant
}
}

```

Three sync channels

Channel	Carries	Watch out
AVPlayer .playback Coordinator (15) GroupSession Messenger (15)	rate, time, pause, seek — intercepted and imposed on the others your Codable/Data to .all or .only(p); ≤ 256 KB ; .reliable (default) or .unreliable (16, UDP)	does not manage the queue; stalls/interruptions <i>suspend</i> coordination live traffic only — no history for late joiners
GroupSession Journal (17)	Transferable files; add, attachments, load	available to everyone <i>whenever</i> they joined; not for fast messages

Local detour: let s =
player.playbackCoordinator.beginSuspension(for: ...) — rewind
2x for yourself, s.end() rejoins the group's time. Same movie,
different URLs per device (CDN, locale)?
AVPlayerPlaybackCoordinatorDelegate
playbackCoordinator(_:identifierFor:) returns a shared id.

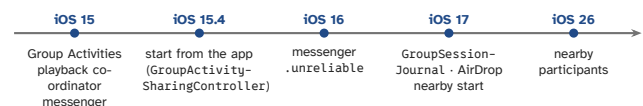
Privacy and platforms

- E2E encrypted; Apple sees neither content nor playback events, nor your app's identity. Participant.id is a UUID, no personal data — map people to accounts yourself.
- iOS 26: Participant.isNearbyWithLocalParticipant (same room vs remote). **visionOS**: spatial Personas, session.systemCoordinator, SpatialTemplate for seating; visionOS 26 shares with people nearby.

Traps

- ! `activate()` with no call/conversation **fails** — check eligibility, fall back to `GroupActivitySharingController` or the share sheet.
- ! The coordinator syncs *time*, not *which item*: drive item changes through `session.activity`, or players drift to different movies.
- ! Late joiner sees an empty canvas: the messenger never replays. Diff `$activeParticipants`, send a snapshot to `.only(new)` — or keep state in the `Journal`.
- ! `.unreliable` is best-effort: cursors and drags, never moves or scores.
- ! `.invalidated` is terminal: drop the session, messenger and tasks.
- `join()` only once your UI can show the activity.
- ! “SharePlay needs my backend” — no: `FaceTime`’s infrastructure relies it; you only need content every device can load.

Timeline



Remember

Activate once, receive everywhere, join when ready, sync three ways (Coordinator · Messenger · Journal), release on invalidated.